

OAKLEY 19 SSW, KS, USA

WMO: 724570

Lat: 38.8701N

Lon: 100.9627W

Elev: 875

StdP: 91.25

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 03067

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.0	-14.2	-21.3	0.6	-14.8	-18.9	0.8	-11.1	11.8	0.9	10.6	-1.0	2.7	N/A	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	37.7	19.7	36.1	19.8	34.4	19.7	22.5	31.2	21.8	30.8	21.2	30.4	4.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	16.6	25.2	19.4	15.7	24.7	18.7	15.0	24.2	70.7	31.4	68.0	30.9	65.7	30.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.1	7.8	-22.0	40.7	3.0	1.7	-24.2	41.9	-26.0	42.9	-27.6	43.8	-29.8	45.1
			-22.9	23.7	3.1	0.8	-25.2	24.2	-27.0	24.7	-28.8	25.1	-31.0	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-0.6	0.2	6.7	11.0	16.5	23.3	25.9	24.4	20.2	12.1	5.5	-0.9
	DBStd	10.82	5.83	6.43	5.91	5.23	4.96	3.80	3.04	2.95	4.55	5.42	5.34	5.87
	HDD10.0	1332	331	279	134	49	7	0	0	0	1	40	154	337
	HDD18.3	2997	588	509	362	223	94	5	1	1	32	202	386	595
	CDD10.0	2093	1	3	31	80	209	400	492	448	307	106	18	1
	CDD18.3	717	0	0	1	3	37	155	234	190	88	8	0	0
	CDH23.3	8924	1	8	58	152	579	1922	2824	2083	1063	215	18	0
	CDH26.7	4508	0	2	12	41	228	993	1578	1084	500	69	1	0
Wind	WSAvg	3.4	3.1	3.4	3.8	4.4	3.8	3.5	3.1	2.9	3.1	3.3	3.1	3.0
Precipitation	PrecAvg	490	11	11	26	48	72	63	88	65	41	36	17	12
	PrecMax	658	49	41	78	111	161	125	213	132	111	101	69	100
	PrecMin	320	0	0	0	15	32	12	4	13	8	0	0	0
	PrecStd	84	12	10	18	24	35	30	47	33	28	34	18	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	24.3	28.5	30.6	35.2	40.8	39.4	38.8	36.9	31.8	25.7	18.5
		MCWB	8.1	9.8	12.5	12.8	15.9	18.3	20.5	19.7	18.3	15.9	11.5	7.7
	2%	DB	15.6	19.1	24.8	27.5	31.6	36.9	37.8	36.4	34.4	28.6	22.5	15.0
		MCWB	6.2	7.9	11.3	13.1	17.1	19.1	20.3	20.3	18.4	14.5	10.6	5.9
	5%	DB	12.3	14.8	21.3	24.8	29.2	34.6	36.3	34.6	32.2	25.7	18.7	11.8
		MCWB	4.8	6.0	9.7	12.4	16.4	19.5	20.6	19.9	18.3	13.4	8.7	5.2
	10%	DB	8.7	11.0	17.8	21.4	26.6	32.3	34.6	32.6	29.6	22.4	15.1	8.5
		MCWB	3.1	4.2	8.7	11.5	15.9	19.4	20.5	19.9	17.8	12.2	7.3	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.2	15.4	16.2	20.7	22.9	23.6	23.2	21.5	18.1	12.9	9.6
		MCDB	17.7	20.3	22.9	24.3	28.2	31.0	32.2	31.4	30.5	24.6	19.5	13.5
	2%	WB	6.5	8.2	12.5	15.0	19.2	21.8	22.7	22.2	20.4	16.1	11.3	6.7
		MCDB	14.5	19.0	20.7	22.8	26.2	30.8	31.6	30.9	29.2	24.2	19.1	13.1
	5%	WB	4.9	6.4	10.8	13.7	18.0	21.0	21.9	21.5	19.6	14.6	9.6	5.0
		MCDB	11.5	14.7	19.6	21.0	25.6	30.3	31.1	30.3	28.4	23.0	18.2	11.7
	10%	WB	3.3	4.4	9.2	12.3	16.8	20.2	21.3	20.9	18.9	13.1	7.7	3.2
		MCDB	8.2	10.8	18.1	20.2	24.8	29.4	30.9	29.5	27.1	20.1	15.0	8.5
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.9	16.5	15.5	14.9	15.3	14.7	14.5	15.2	15.8	16.3	14.2
		MCDBR	20.7	22.2	22.4	21.0	18.6	18.2	17.2	17.6	18.1	20.9	21.7	19.8
		MCWBR	12.2	12.2	11.1	9.4	7.4	5.6	4.7	5.5	6.0	9.2	11.4	12.1
	5% WB	MCDBR	19.1	21.7	20.5	16.8	15.8	15.8	15.1	14.8	15.2	17.4	19.9	18.2
		MCWBR	11.7	12.4	10.8	8.6	7.5	6.3	5.0	5.3	5.5	8.5	10.6	11.4
Clear-Sky Solar Irradiance	taub	0.260	0.267	0.297	0.324	0.351	0.371	0.391	0.368	0.340	0.296	0.267	0.257	
	taud	2.545	2.547	2.469	2.419	2.411	2.388	2.376	2.402	2.459	2.564	2.567	2.552	
	Ebn at Noon	939	974	968	953	927	904	883	897	907	926	921	916	
	Edh at Noon	73	83	100	113	116	119	120	114	101	81	71	67	
All-Sky Solar Radiation	RadAvg	2.56	3.38	4.66	5.71	6.60	7.32	7.25	6.30	5.22	3.82	2.82	2.25	
	RadStd	0.21	0.31	0.39	0.30	0.41	0.35	0.26	0.35	0.30	0.39	0.22	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page